1 2

1 Home_Die_Motor

2 LatchHomeDie PolyPipe.Ax1In3

2.1

3 LatchHomeDie SystemEnable

4 LatchHomeDie SystemEnable

5 Home_Table_Motor

6 LatchHomeTable PotentiometerPositionMM < Potent_Edge_scale_mm

7 LatchHomeTable PotentiometerPositionMM > Potent_Edge_scale_mm

3 4 5 6 7 8 9 10

1 >

2 >

2.1 >

3 >

4 >

5 >

6 >

SystemEnable

7 >

1 > LatchHomeDie (SET)

2 > LatchHomeDie (RST)

2.1 > Enable ↑
SET POSITION
Module Name: AMC-1
Channel/Axis: 1 Channel/Axis Name: Die Motor
Set Position To 0

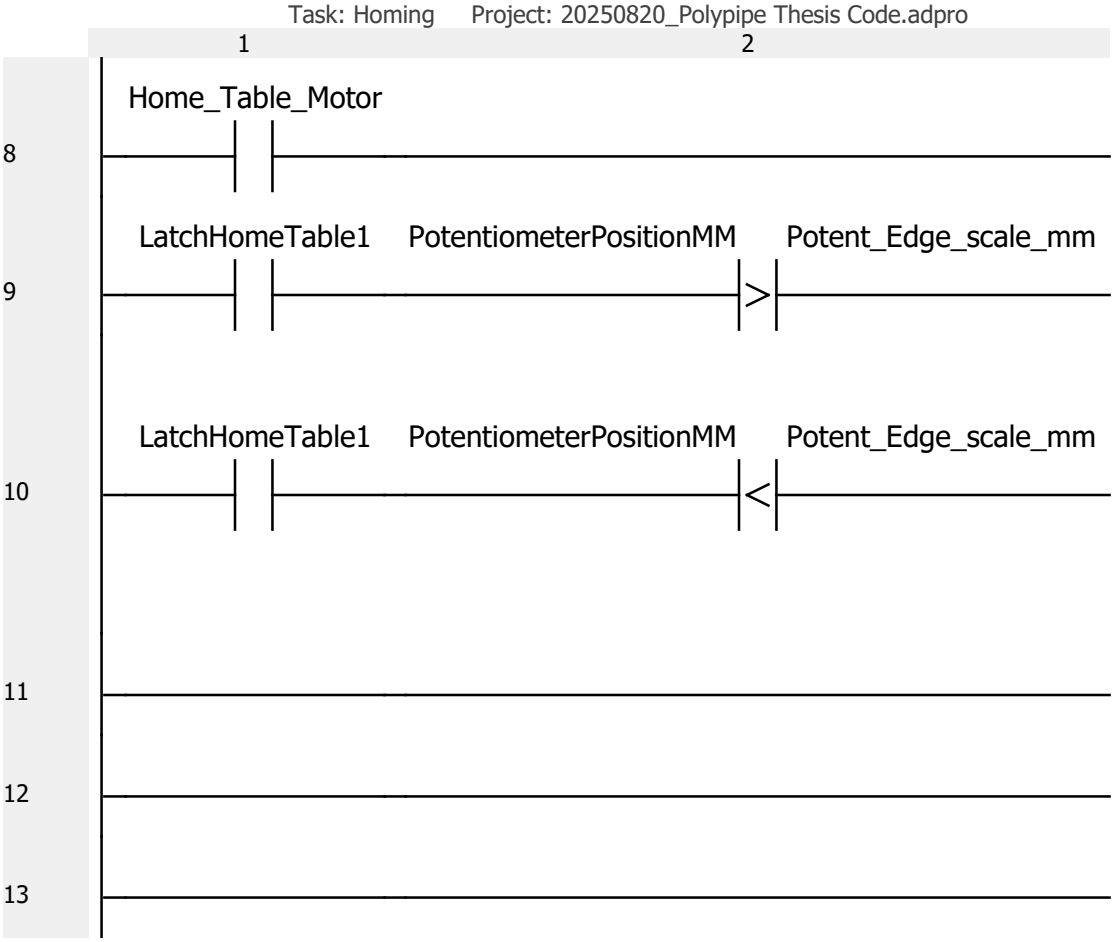
3 > Enable
VELOCITY MOVE
Module Name: AMC-1
Channel/Axis: 1 Channel/Axis Name: Die Motor
Move Direction Positive
Velocity 0.5
Ramp Rate 1
Stop Setup Immediate Stop

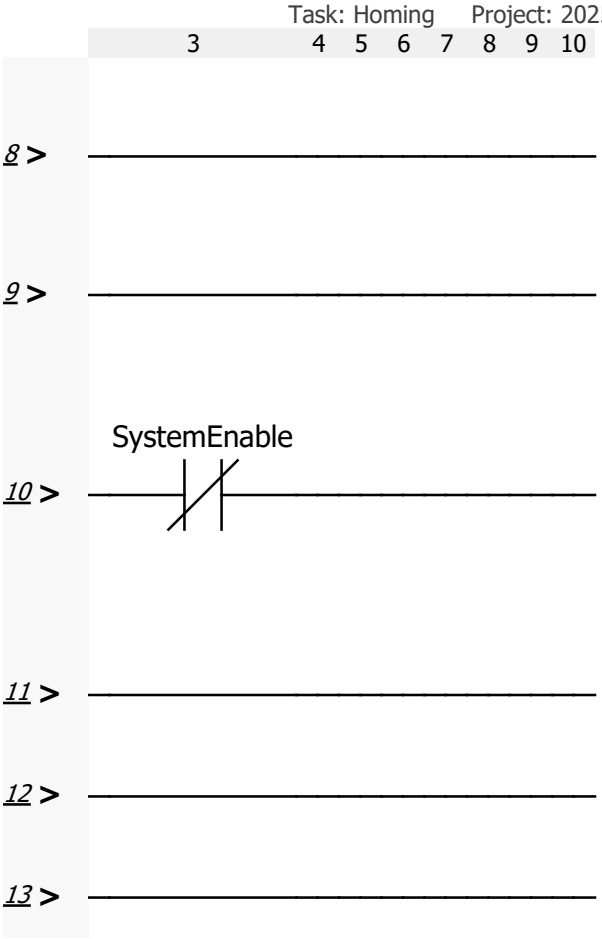
4 > Enable
VELOCITY MOVE
Module Name: AMC-1
Channel/Axis: 2 Channel/Axis Name: Top Motor
Move Direction Negative
Velocity 2
Ramp Rate 4
Stop Setup Immediate Stop

5 > LatchHomeTable (SET)

6 > LatchHomeTable (RST)

7 > Enable
VELOCITY MOVE
Module Name: AMC-1
Channel/Axis: 3 Channel/Axis Name: Table Motor
Move Direction Negative
Velocity 3
Ramp Rate 1
Stop Setup Immediate Stop


1
2
8
Home_Table_Motor
9
LatchHomeTable1
PotentiometerPositionMM
>
Potent_Edge_scale_mm
10
LatchHomeTable1
PotentiometerPositionMM
<
Potent_Edge_scale_mm
11
12
13


3 4 5 6 7 8 9 10
8 >
9 >
SystemEnable
10 >
11 >
12 >
13 >

11

8 > LatchHomeTable1 (SET)

9 > LatchHomeTable1 (RST)

10 > Enable

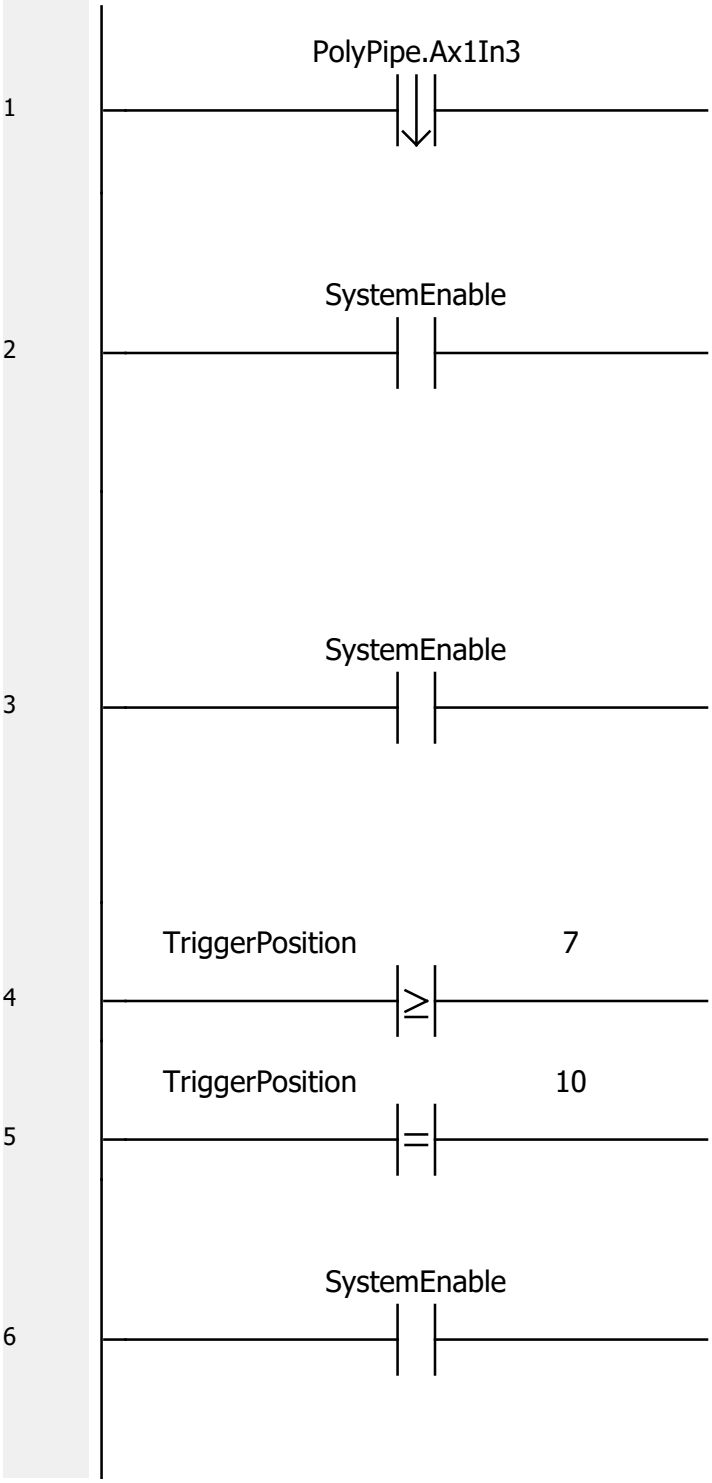
VELOCITY MOVE
Module Name: AMC-1
Channel/Axis: 3 Channel/Axis Name: Table Motor
Move Direction Positive
Velocity 3
Ramp Rate 1
Stop Setup Immediate Stop

11 > (END)

12 > (END)

13 > (END)

1


PolyPipe.Ax1In3
1
SystemEnable
2
SystemEnable
3
TriggerPosition
7
4
TriggerPosition
10
5
SystemEnable
6

2 3

1 >

2 >

3 >

TriggerPosition 9

4 > ≤

5 >

PotentiometerPositionMM Die Index Position1 CV 25

6 > > >

4 5 6 7 8 9 10

1 >

2 >

3 >

4 >

5 >

In_Disable_Range

6 >

11

1 >

Enable

COPY DATA
Source Destination
0 PolyPipe.Ax1Position
Type Copy Value

2 >

Enable

Top Motor - VELOCITY MOVE
Module Name: AMC-1 In Progress InProg
Channel/Axis: 2 Channel/Axis Name: Top Motor Complete Complet
Move Direction Negative Move Status Statu
Velocity Velocity
Ramp Rate RampRate
Stop Setup Stop at Maximum Decel Rate

3 >

Enable

SIMPLE MOVE
Module Name: AMC-1
Channel/Axis: 1 Channel/Axis Name: Die Motor
Move Addressing Absolute
Direction Negative
Target 1
Velocity PolyPipe.Ax4Velocity
Acceleration DieAccelDecelRate
Deceleration DieAccelDecelRate
Stop Setup Immediate Stop

4 >

In_Disable_Range
(SET)

5 >

In_Disable_Range
(RST)

6 >

Enable

VELOCITY MOVE
Module Name: AMC-1
Channel/Axis: 3 Channel/Axis Name: Table Motor
Move Direction Negative
Velocity Table Motor Velocity
Ramp Rate Table Motor Ramp Rate
Stop Setup Immediate Stop

1 >

gress
:e
2 > s

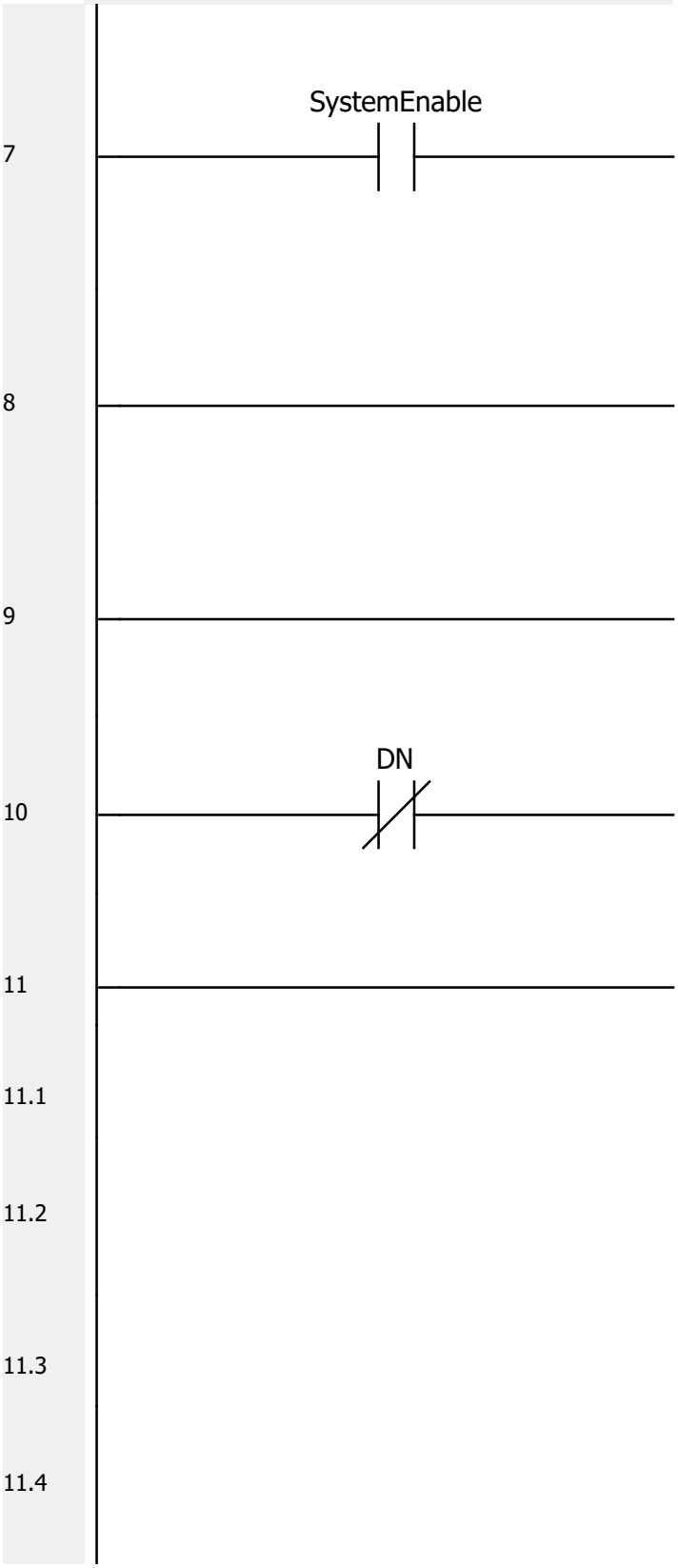
3 >

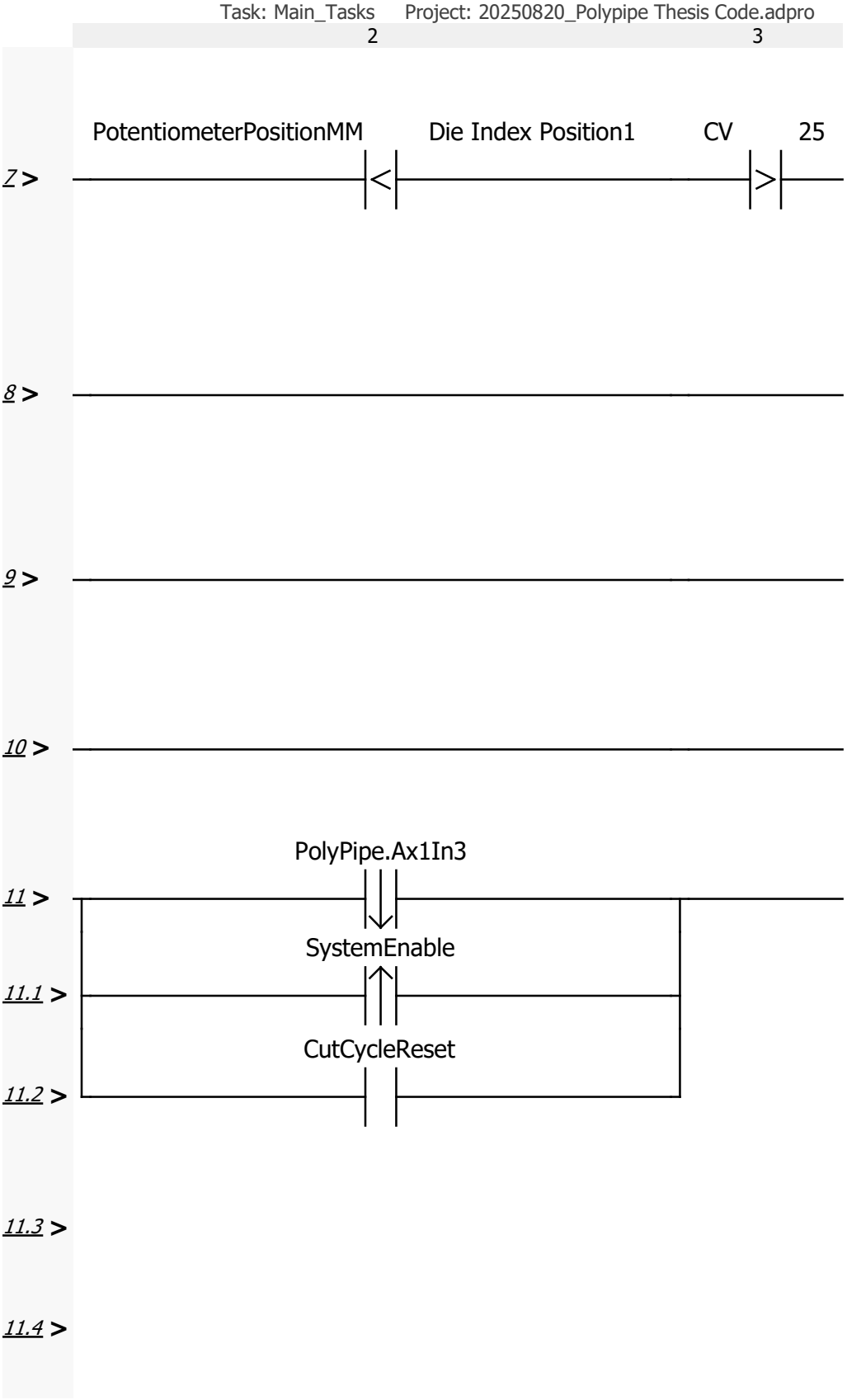
4 >

5 >

6 >

1




2
3
PotentiometerPositionMM
Die Index Position1
CV
25
7 >
8 >
9 >
10 >
PolyPipe.Ax1In3
11 >
SystemEnable
11.1 >
CutCycleReset
11.2 >
11.3 >
11.4 >

4 5 6 7 8 9 10

7 >

In_Disable_Range

8 >

9 >

10 >

11 >

11.1 >

SystemEnable

11.2 >

11.3 >

11.4 >

11

7 > Enable

VELOCITY MOVE
Module Name: AMC-1
Channel/Axis: 3 Channel/Axis Name: Table Motor
Move Direction Positive
Velocity Table Motor Velocity
Ramp Rate Table Motor Ramp Rate
Stop Setup Immediate Stop

8 > Enable

MATH
Result Table Motor Velocity
Numeric Result
Formula
abs(Die Index Position1- PotentiometerPositionMM) * 5

9 > Enable

MATH
Result Table Motor Ramp Rate
Numeric Result
Formula
Table Motor Velocity * 1.2 * 5

10 > Time Up

SIMPLE TIMER
Preset Value 150 Current Value CV
On Delay Done DN
Time Base .001 sec

11 > StopCutCycle (SET)

11.1 > TriggerCut (RST)

11.2 > Enable

SET POSITION
Module Name: AMC-1
Channel/Axis: 4 Channel/Axis Name: Encoder
Set Position To 0

11.3 > TriggerCutAtPosition (RST)

11.4 > Enable

SET POSITION
Module Name: AMC-1
Channel/Axis: 1 Channel/Axis Name: Die Motor
Set Position To 0

1

12 TriggerCut

13 Row Spacing Inches = 38

14 Row Spacing Inches = 40

15 Row Spacing Inches = 76

16 TriggerCutAtPosition

17 TriggerCut

18 SystemEnable

2 3

12 >

13 >

14 >

15 >

16 >

17 > StopCutCycle SystemEnable

18 >

4 5 6 7 8 9 10

12 >

13 >

14 >

15 >

16 >

17 >

18 >

StopCutCycle

12 > (RST)

13 > Enable

MATH
Result TriggerPosition
Numeric Result
Formula
PolyPipe.Ax4EncPosition / 29.29232081

14 > Enable

MATH
Result TriggerPosition
Numeric Result
Formula
PolyPipe.Ax4EncPosition / 29.21018583

15 > Enable

MATH
Result TriggerPosition
Numeric Result
Formula
PolyPipe.Ax4EncPosition / 28.97615132

TriggerCut

16 > (SET)

17 > Enable

GEAR DRIVETRAIN
Module Name: AMC-1
Drivetrain Segments 1
Encoder as Primary Master Axis Yes
Primary
Master Encoder Slave Die Motor
Ratio 40.3299 Disable

18 > Enable

GEAR DRIVETRAIN
Module Name: AMC-1
Drivetrain Segments 1
Encoder as Primary Master Axis Yes
Primary
Master Encoder Slave Top Motor
Ratio -40.3299 Disable

1

19

20

21

2 3

19 >

20 >

21 >

4 5 6 7 8 9 10

19 >

20 >

21 >

11

19 > Enable

SCALE NON-LINEAR

Input Edge Sensor Analog Output Edge Position MM

Input Value	Desired Output
11471	0
13210	.06
15694	1.48
21391	4.74
26289	7.55
32208	10.94
39272	14.98
45110	18.33
52250	22.42
59139	26.36
65160	29.82
65565	30

20 > Enable

SCALE NON-LINEAR

Input R_edge sensor Output R_Edge Position MM

Input Value	Desired Output
11471	0
13210	.06
15694	1.48
21391	4.74
26289	7.55
32208	10.94
39272	14.98
45110	18.33
52250	22.42
59139	26.36
65160	29.82
65565	30

21 > Enable

SCALE LINEAR

Input PolyPipe.Ax2EncPosition Output Cutting Motor Degrees

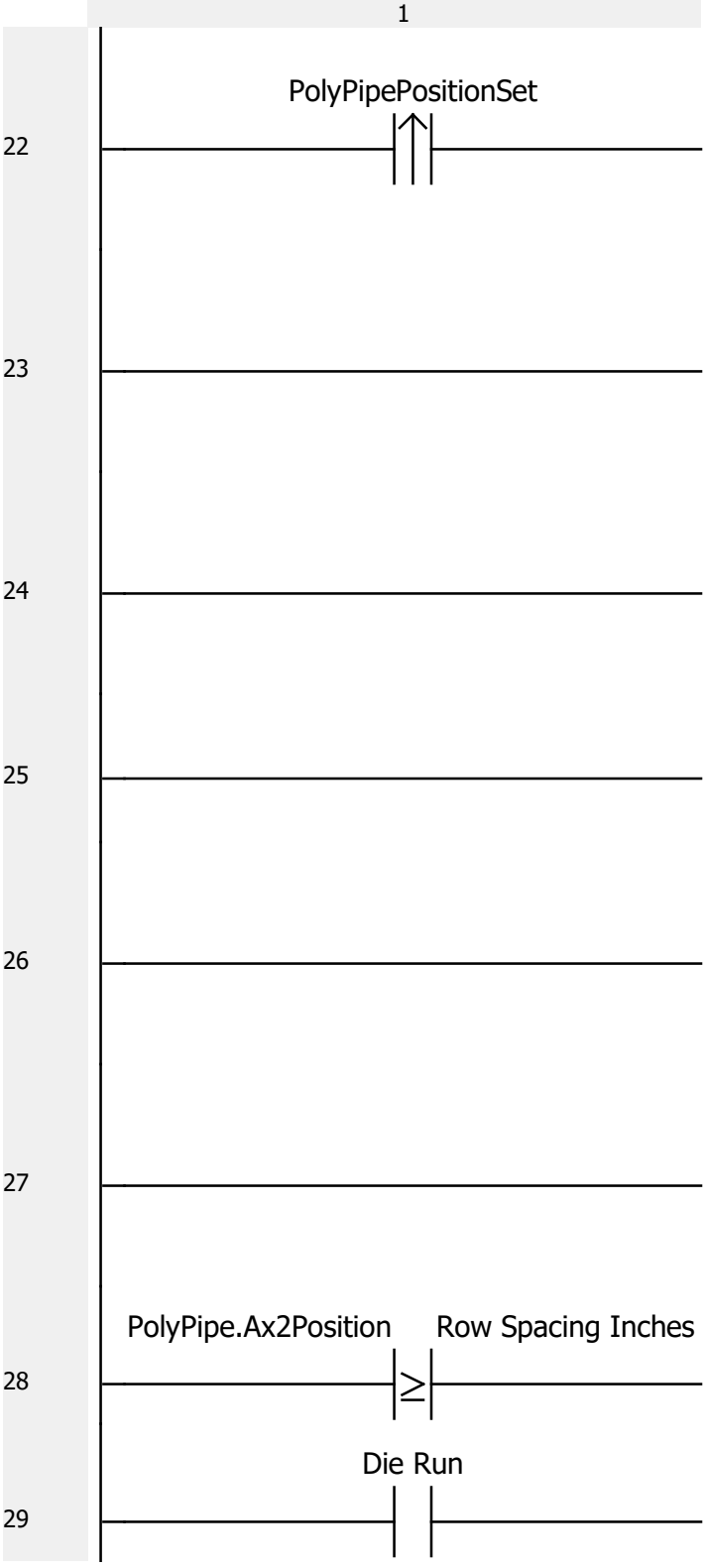
In Min 0

In Max 400

Out Min 360

Out Max 0

1


PolyPipePositionSet
22
23
24
25
26
27
PolyPipe.Ax2Position
Row Spacing Inches
28
Die Run
29

2 3

22 >

23 >

24 >

25 >

26 >

27 >

28 >

29 >

4 5 6 7 8 9 10

22 >

23 >

24 >

25 >

26 >

27 >

28 >

29 >

22 > Enable

COPY DATA
Source: PolyPipe.Ax4EncPosition → Destination: PolyPipePositionCapture
PolyPipe.Ax2EncPosition → Die Position Capture
Type: Copy Value

23 > Enable

MATH
Result: PolyPipePunchTrigger
Numeric Result
Formula
PolyPipePositionCapture + ((Row Spacing Inches / 6.31974025) * 360

24 > Enable

MATH
Result: Avg_Edge_Position
Numeric Result
Formula
(Edge Position MM + R_Edge Position MM) / 2

25 > Enable

COMPARE VALUES
Input 1: PolyPipe.Ax4EncPosition Equal To PolyPipePositionSet
Input 2: PolyPipePunchTrigger

26 > Enable

MATH
Result: DieAccelDecelRate
Numeric Result
Formula
PolyPipe.Ax4Velocity ^2

27 > Enable

MATH
Result: Die Position Target
Numeric Result
Formula
Die Position Capture + 400

Die Run
28 > (OUT)

Cutting Motor Off
29 > (OUT)

22 >

23 >

)

24 >

25 >

26 >

27 >

28 >

29 >

1

30

31 Diameter_MM 0 =

32 Diameter_MM 1.587 ≥

33 Diameter_MM 3.17 ≥

34 Diameter_MM 4.76 ≥

35 Diameter_MM 6.3 ≥

36 Diameter_MM 7.93 ≥

37 Diameter_MM 9.52 ≥

2 3

30 >

31 >

32 > Diameter_MM ≤ 1.588

33 > Diameter_MM ≤ 3.18

34 > Diameter_MM ≤ 4.77

35 > Diameter_MM ≤ 6.4

36 > Diameter_MM ≤ 7.94

37 > Diameter_MM ≤ 9.53

4 5 6 7 8 9 10

30 >

31 >

32 >

33 >

34 >

35 >

36 >

37 >

30 > Enable

MATH
Result Diameter_MM
Numeric Result
Formula
(DiameterIn * 25.4)

31 > Enable

COPY DATA
Source Destination
0 Hole Size
Type Copy Value

32 > Enable

COPY DATA
Source Destination
0.306510974 Hole Size
Type Copy Value

33 > Enable

COPY DATA
Source Destination
0.751371601 Hole Size
Type Copy Value

34 > Enable

COPY DATA
Source Destination
1.293767897 Hole Size
Type Copy Value

35 > Enable

COPY DATA
Source Destination
1.918095495 Hole Size
Type Copy Value

36 > Enable

COPY DATA
Source Destination
2.650424332 Hole Size
Type Copy Value

37 > Enable

COPY DATA
Source Destination
3.347503545 Hole Size
Type Copy Value

1

38 Diameter_MM ≥ 11.11

39 Diameter_MM ≥ 12.6

40 Diameter_MM ≥ 14.28

41 Diameter_MM ≥ 15.87

42 Diameter_MM ≥ 17.46

43 Diameter_MM ≥ 19.0

44 Diameter_MM ≥ 20.63

2 3

38 > Diameter_MM ≤ 11.12

39 > Diameter_MM ≤ 12.8

40 > Diameter_MM ≤ 14.29

41 > Diameter_MM ≤ 15.88

42 > Diameter_MM ≤ 17.47

43 > Diameter_MM ≤ 19.1

44 > Diameter_MM ≤ 20.64

4 5 6 7 8 9 10

38 >

39 >

40 >

41 >

42 >

43 >

44 >

11

38 > Enable
COPY DATA
Source Destination
4.180922369 Hole Size
Type Copy Value

39 > Enable
COPY DATA
Source Destination
5.081588517 Hole Size
Type Copy Value

40 > Enable
COPY DATA
Source Destination
5.957857812 Hole Size
Type Copy Value

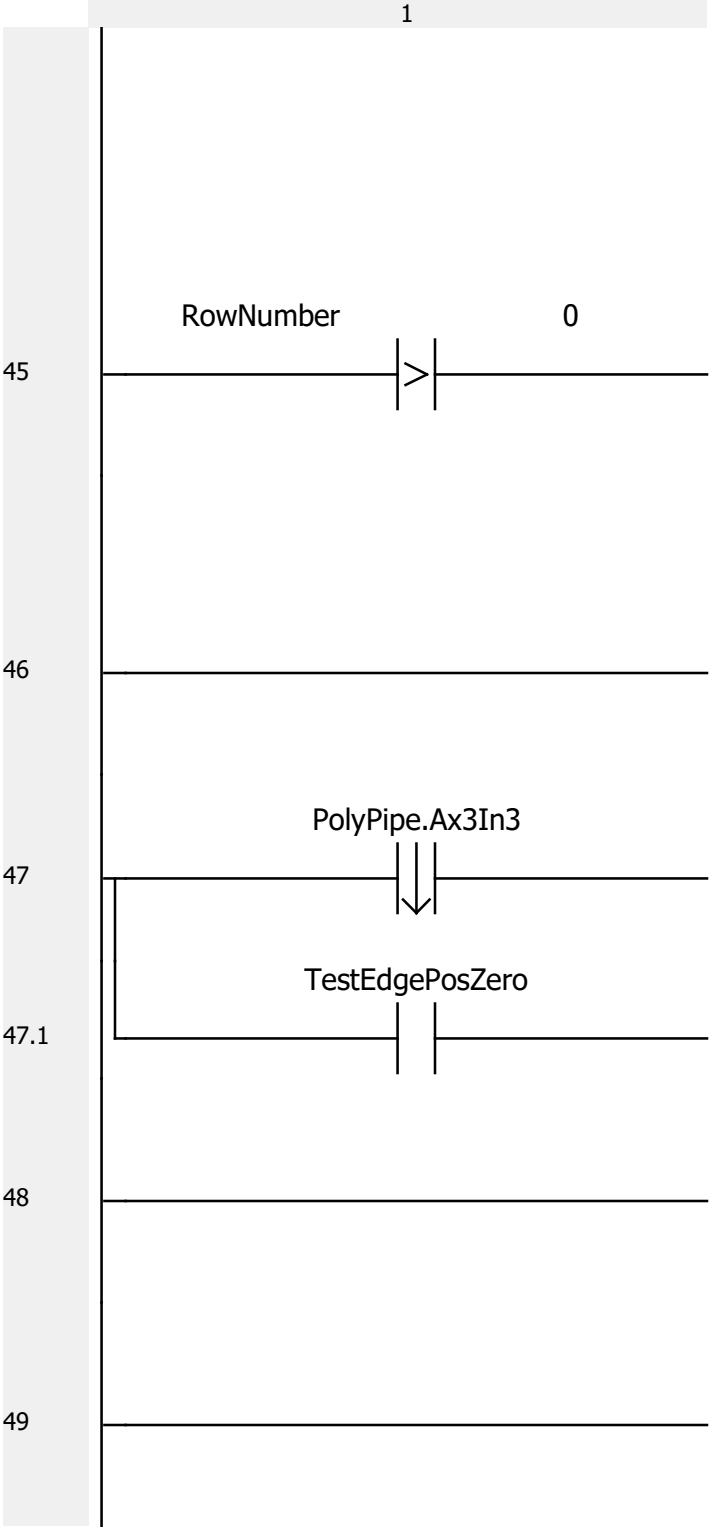
41 > Enable
COPY DATA
Source Destination
6.959293606 Hole Size
Type Copy Value

42 > Enable
COPY DATA
Source Destination
7.997536308 Hole Size
Type Copy Value

43 > Enable
COPY DATA
Source Destination
9.150634338 Hole Size
Type Copy Value

44 > Enable
COPY DATA
Source Destination
10.31875 Hole Size
Type Copy Value

1


RowNumber
0
45
46
PolyPipe.Ax3In3
47
TestEdgePosZero
47.1
48
49

2 3

45 >

46 >

47 >

47.1 >

48 >

49 >

4 5 6 7 8 9 10

45 >

46 >

47 >

47.1 >

48 >

49 >

This function is to find the desired hole size that needs to be cut. Hole Size take into account the area of water flow desired for that row. this is added against the potentiometer which should always be searching for the zero value of the material. There is an offset value added to compensate for the true zero position as it is hard to find true zero by eyeballing it.

45 > Enable

MATH
Result Die Index Target
Numeric Result
Formula
Hole Size + Potent_Edge_scale_mm + 1.809375

This is just for reference. I wanted to see the difference the Error Correction equation is having on the movement of the die.

46 > Enable

MATH
Result Die Index Position1
Numeric Result
Formula
Hole Size + Potent_Edge_scale_mm + 1.809375

47 > Enable

COPY DATA

Source	Destination
Avg_Edge_Position	Edge Position Reference Zero

Type Copy Value

47.1 >

48 > Enable

MATH
Result Die Index Velocity
Numeric Result
Formula
abs(PolyPipe.Ax3Position - Die Index Position1)

49 > Enable

MATH
Result Die Index Ramp Rate
Numeric Result
Formula
Die Index Velocity^2

	1
50	
51	
52	

2 3

50 >

51 >

52 >

4 5 6 7 8 9 10

50 >

51 >

52 >

11

Reference for die travel. Out min and Out Max are in mm. Gefran potentiometer linear position sensor

50 >

```
Enable ──┤ SCALE LINEAR
           Input  Die Table Position Analog  Output  PotentiometerPositionMM
           In Min  0
           In Max  65535
           Out Min  0
           Out Max  50.8
```

Not sure where the values are coming from yet. However, this is taking the 'Edge Position MM' which is the Edge Sensors analog signal converted to mm. Then it inverts the data so that '+' direction is the same as the '+' direction on the Potentiometer connected to the table.
This allows us to index the die.

51 >

```
Enable ──┤ SCALE LINEAR
           Input  Avg_Edge_Position  Output  Potent_Edge_scale_mm
           In Min  0.0
           In Max  30.0
           Out Min  30.92
           Out Max  0.92
```

52 > ──(END)

1 2

1 TriggerPosition Row Spacing Inches =

2 TriggerPosition Row Spacing Inches >

2.1

3

4 TriggerCutAtPosition

4.1

4.2 CutCycleReset

5 CutCycleReset

5.1

5.2

5.3

5.4

3 4 5 6 7

1 >

Row Spacing Inches > 0 SystemEnable

2 >

Row Spacing Inches > 0 SystemEnable

2.1 >

3 >

4 >

4.1 >

4.2 >

5 >

5.1 >

5.2 >

5.3 >

5.4 >

	8	9
1 >		
2 >		
2.1 >		
3 >		
4 >		
4.1 >		
4.2 >		
5 >		
5.1 >		
5.2 >		
5.3 >		
5.4 >		

10

1 >

2 >

2.1 >

3 >

4 >

4.1 >

4.2 >

5 >

5.1 >

5.2 >

5.3 >

5.4 >

11

TriggerCutAtPosition

1 > —(SET)

2 > Enable

SET POSITION
Module Name: AMC-1
Channel/Axis: 4 Channel/Axis Name: Encoder
Set Position To 0.1428571429

TriggerCutAtPosition

2.1 > —(RST)

3 > Enable

MATH
Result HolePercentComplete
Numeric Result
Formula
(RowNumber /Total_Row_Number)*100

4 > Cnt Up

4.1 > Cnt Down

4.2 > Reset

COUNTER
Preset Value Total_Row_Number Current Value RowNumber
Don't Reset

Light1

5 > —(RST)

Light2

5.1 > —(RST)

Light3

5.2 > —(RST)

Light4

5.3 > —(RST)

Light5

5.4 > —(RST)

1 2

5.5

5.6

5.7

5.8

5.9

5.10

5.11

5.12

6 RowNumber ≤ 0

6.1

7 RowNumber > 0

7.1

3 4 5 6 7

5.5 >

5.6 >

5.7 >

5.8 >

5.9 >

5.10 >

5.11 >

5.12 >

6 >

6.1 >

RowNumber RowCount

7 > ≤

7.1 >

8 9

5.5 >

5.6 >

5.7 >

5.8 >

5.9 >

5.10 >

5.11 >

5.12 >

6 >

RowNumber 0

6.1 >

7 >

7.1 >

10

5.5 >

5.6 >

5.7 >

5.8 >

5.9 >

5.10 >

5.11 >

5.12 >

6 >

TriggerPosition 9

6.1 > [>]

7 >

SystemEnable

7.1 > [ ]

Light6

5.5 > (RST)

Light7

5.6 > (RST)

Light8

5.7 > (RST)

Light9

5.8 > (RST)

Light10

5.9 > (RST)

Light11

5.10 > (RST)

Light12

5.11 > (RST)

Light13

5.12 > (RST)

6 > Enable

COPY DATA
Source: Avg_Edge_Position
Destination: Die Index Position1
Type: Copy Value

6.1 > Enable

COPY DATA
Source: HoleS1
Destination: DiameterIn
Type: Copy Value

7 > Enable

COPY DATA
Source: HoleS1
Destination: DiameterIn
Type: Copy Value

Light1

7.1 > (SET)

1 2

7.2

8 RowNumber > RowCount

8.1

8.2

8.3

9 RowNumber > Total_Row_Number

10 RowNumber > RowCount6

11

12

13

14

3 4 5 6 7

7.2 >

RowNumber RowCount2

8 > ≤

8.1 >

8.2 >

8.3 >

9 >

10 >

11 >

12 >

13 >

14 >

8 9

7.2 >

RowNumber RowCount

=

8 >

8.1 >

8.2 >

RowNumber RowCount2

8.3 >

=

9 >

10 >

11 >

12 >

13 >

14 >

10

TriggerPosition 9

7.2 >

8 >

8.1 >

8.2 >

TriggerPosition 9

8.3 >

9 >

10 >

11 >

12 >

13 >

14 >

11

7.2 >
Enable
COPY DATA
Source Destination
HoleS2 DiameterIn
Type Copy Value

8 >
Enable
COPY DATA
Source Destination
HoleS2 DiameterIn
Type Copy Value

Light2
8.1 > (SET)

Light1
8.2 > (RST)

8.3 >
Enable
COPY DATA
Source Destination
HoleS3 DiameterIn
Type Copy Value

9 >
Enable
COPY DATA
Source Destination
0 DiameterIn
Type Copy Value

Light6
10 > (RST)

11 > (END)

12 > (END)

13 > (END)

14 > (END)

I/O Modules

Part Number	Group Number	Base Number	Slot Number	Description
P2-550	0	1	0	CPU Module
P2-06TRS	0	1	1	6PT, ISOLATED OUTPUT RELAY
P2-04AD	0	1	2	4CH, 16-BIT, VOLTAGE/CURRENT, ANALOG INPUT
P2-08SIM	0	1	3	8PT, INPUT SIMULATOR
P2-SCM	0	1	4	Serial Communication Module

GS Drives

Part Number	Node Number	Horse Power Rating

Protos X

Module P/N	Group	Position	Module Description

Modbus Devices

Device Name	Port Type	Ethernet Port	IP Address Type	IP Address	Port No.	Ethernet Slave Node	Enable TCP Connect	Serial Port	Serial Slave Node	TCP Connecte	TCP/IP Error

Data Block	I/O T...	Enabl...	Tag Name	Data Type	Element Count	Message Size	Connection Point (I/O)	Class (Explicit)	Instance (Explicit)	Attribute (Explicit)
1	Input	No					101 (0x65)	4 (0x4)	101 (0x65)	3 (0x3)
	Output	No					102 (0x66)	4 (0x4)	102 (0x66)	3 (0x3)

Devices

Device Name	Ethernet Port	IP Address	Port No.	Session Timeout Enable	Session Timeout in secs	Byte Order Swap	TCP Connected	Adapter Name	Vendor ID	TCP/IP Error

Messages

Device Name	Message Name	Enable Bit	RPI in msec	Routing Enabled	Slot No.	Connection Online	General Status	Extended Status	Status Description

Data

Device Name	Message Name	Data Name	Config Enabled	Delivery Option	RPI in msec	Connection Point	Data Array	Element Count (Msg Size)	Status Header Included